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Sales Lead Scoring Tool using AI

Requirements Gathering & System Architecture

1. Project Overview

The AI Lead Scoring Tool is a web-based application designed to assist B2B sales teams in qualifying and prioritising inbound leads using artificial intelligence. Rather than relying on static, rule-based point systems, the tool leverages a Large Language Model (GPT-OSS 120B via Groq) to reason holistically across multiple lead attributes and generate an actionable score, tier classification, risk analysis, and CRM-ready note in a single API call. The frontend is built with Streamlit, a Python-native UI framework that transforms a single script into a deployable web application without requiring separate HTML, CSS, or JavaScript files.

2. Requirements Gathering & Field Selection Rationale

The form fields were not chosen arbitrarily. Each input maps directly to a well-established lead qualification criterion supported by industry research and sales methodology. The following sections document the research basis for each field group.

2.1 Firmographic Fields — Company Name, Industry, Size, Revenue

Firmographic data represents the organisational attributes of a lead, the equivalent of demographics for a business. According to ZoomInfo's lead scoring methodology guide, firmographic criteria are the foundation of any B2B scoring model, used to determine whether a company matches the Ideal Customer Profile (ICP) in terms of scale, sector, and financial capacity (ZoomInfo, 2025).

Company Name is collected as the anchor identifier for the lead record and for CRM note generation. Industry is captured because, as research by Leads at Scale confirms, over 50% of B2B companies weight industry as a primary scoring criterion, different verticals carry different buying patterns, procurement cycles, and product fit (Leads at Scale, 2026). A SaaS company in the Technology sector will respond differently to CRM solutions than a company in Manufacturing.

Company Size and Annual Revenue are captured together because they jointly validate whether a prospect has the infrastructure and financial capacity to buy. As the Mojenta B2B Tech Scoring Guide states, a startup with five employees has fundamentally different procurement capability than an enterprise with 5,000 workers, even if their stated interest is identical (Mojenta, 2025). Revenue also provides a proxy for budget availability, one of the four pillars of the BANT qualification framework (see Section 2.3).

2.2 Contact Fields - Contact Name, Job Title

Contact-level data captures the authority dimension of lead qualification. The job title field is one of the highest-weighted criteria in B2B scoring because purchasing authority correlates directly with seniority. Research from Leads at Scale assigns C-level executives +30 points, VPs/Directors +25 points, and Managers +15 pointsm reflecting the reality that a CEO exploring your pricing page carries far more conversion weight than a junior employee doing the same (Leads at Scale, 2026).

As further corroborated by ZoomInfo's scoring model documentation, the key question the job title field answers is: 'Is this person a decision-maker, influencer, or end user?', three roles that require entirely different sales engagement strategies (ZoomInfo, 2025). Selling to someone without authority, as the Cleverly B2B guide illustrates with a cautionary example, leads to stalled deals and wasted SDR time (Cleverly, 2025).

2.3 Engagement Level

The Engagement Level field captures behavioural intent signals, the implicit dimension of lead scoring that reveals what a prospect is doing, not just who they are. Behavioural scoring distinguishes passive interest from active buying intent. The four-dropdown options map to escalating tiers of commitment:

- Cold (no prior contact) - no engagement signal, lowest intent
- Warm (email opens / webinar) - mid-funnel research behaviour
- Hot (demo booked / call requested) -high-intent action; demo requests score highest in most B2B models
- Existing customer (upsell) - relationship already established, highest baseline trust

According to the Outfunnel Complete Lead Scoring Guide, a lead who has booked a demo is making an explicit time investment that places them far above casual email openers in terms of conversion probability (Outfunnel, 2025). The Belkins scoring system similarly weights demo requests and direct sales inquiries as maximum-point signals (Belkins, 2025).

2.4 Budget Signal

Budget is the 'B' in BANT, the IBM-originated lead qualification framework that has governed B2B sales for over 70 years (Budget, Authority, Need, Timeline). As Salesforce's own BANT documentation explains, budget qualification asks: does the prospect have or can they allocate funds for this solution? (Salesforce, 2024). Without budget confirmation, even a highly engaged, senior decision-maker represents a deal at risk.

The four options in the Budget Signal dropdown capture the full spectrum from no signal through to a competitor evaluation, which signals both budget existence and urgency. The Mojenta guide notes that any indication of budget availability, whether from direct conversation, RFP processes, or indirect signals like funding announcements, should factor directly into the lead score (Mojenta, 2025).

2.5 Pain Points / Notes (Free Text)

The free-text Notes field captures the 'Need' dimension of BANT, the qualitative context that structured dropdowns cannot encode. This field is passed verbatim into the AI prompt, enabling the LLM to reason about industry-specific pain points, stated deadlines, competitive context, and urgency signals that a rules-based scoring system would miss entirely.

The SaaS Hero enterprise scoring guide highlights that need qualification must go beyond binary yes/no — it should capture whether the pain is strategic versus tactical, urgent versus exploratory, and whether the prospect has recognised the problem themselves (SaaS Hero, 2026). The free-text field accommodates all of these nuances.

2.6 Purchase Timeline

Timeline is the fourth pillar of BANT and captures urgency. As Gartner's BANT framework analysis notes, 47% of B2B software purchases take six to nine months to finalise, while 16% close in under three months, meaning a prospect with an 'Immediate' timeline requires a fundamentally different engagement strategy than one exploring options over a year-long horizon (Gartner, 2023). The five dropdown options are ordered from least to most urgent, enabling the LLM to weight timeline signals appropriately in its scoring reasoning.

3. Output Design

3.1 The 0–100 Score with Tier Classification

A numerical score on a 0–100 scale was chosen because it provides a shared, objective language between sales and marketing teams. The Vanderbuild B2B scoring guide notes that a well-calibrated threshold — for example, 75+ points routing directly to Sales, is what transforms a scoring model from a vanity metric into an operational workflow (Vanderbuild, 2025). The three-tier system (Hot / Warm / Cold) maps directly to standard MQL-to-SQL pipeline stages used in Salesforce and HubSpot.

3.2 Strengths and Risks

Rather than outputting a raw score alone, the tool surfaces explicit strengths and risks as discrete items. The Belkins scoring methodology documentation emphasises that the value of a scoring system is not just the number but the confidence it gives a rep, knowing why a lead is scored highly is as important as knowing the score itself (Belkins, 2025). Presenting risks alongside strengths guards against over-reliance on high scores and gives the rep an actionable warning, not just a green light.

3.3 Recommended Next Actions

Three ordered next actions are generated per lead because the output needs to be immediately usable, not just informational. The Monday.com lead scoring research highlights that scoring systems only deliver value when they trigger specific rep behaviours, the score is the triage, the actions are the treatment (Monday.com, 2026). Generating the actions as part of the AI response eliminates the gap between knowing a lead is hot and knowing what to do about it.

3.4 CRM Note

The CRM Note outputs a single, copy-paste-ready sentence for direct entry into Salesforce, HubSpot, or any CRM record. This was included because, as Salesforce's own research cited in the Salesmotion BANT guide notes, sales reps spend only 28% of their time actually selling, administrative tasks including CRM data entry consume a disproportionate share of selling time (Salesmotion, 2026). Generating the note automatically represents a direct, quantifiable time saving per lead.

4. Technology Stack

Streamlit (UI Layer)

Streamlit converts a single Python script into a fully interactive web application. It manages all widget state, layout, and re-rendering without requiring a separate frontend framework. Deployed via Streamlit Community Cloud at share.streamlit.io.

Groq API + GPT-OSS 120B (AI Engine)

Groq provides ultra-low latency inference for open-source LLMs. GPT-OSS 120B is used as the reasoning model. The free tier supports development and light production usage. The model is prompted to return structured JSON, enabling deterministic parsing of AI output.

Python / JSON (Backend Logic)

All data collection, prompt construction, API communication, JSON parsing, and result rendering are handled in a single Python file. No database, no server infrastructure, and no separate deployment pipeline are required.

5. System Architecture - Data Flow

The system operates as a five-stage pipeline from user input to displayed output:

- **Stage 1 : Input Collection:** The user populates the Streamlit form. Each widget stores its value in Streamlit session state. No data is written to disk or a database.
- **Stage 2 : Prompt Construction:** On button click, Python assembles a structured prompt string that embeds all form values and instructs the LLM to return only a valid JSON object in a defined schema.
- **Stage 3 : API Request:** The Groq Python client sends the prompt to the Groq cloud API via HTTPS POST to /v1/chat/completions, using GPT-OSS 120B at temperature=0.3 for consistent, low-variance output.
- **Stage 4 : Response Parsing:** The raw text response is stripped of any markdown code fences and parsed with json.loads(). A try/except block handles malformed response gracefully.
- **Stage 5 : Output Rendering:** Parsed JSON fields are mapped to Streamlit UI components — the score card, tier badge, strengths/risks pills, action list, and CRM note block.

6. References

- ZoomInfo — What is Lead Scoring? How to Build a Scoring Model
- Leads at Scale — 10 Lead Scoring Criteria for B2B Sales
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- Salesforce — What is BANT? The Way to Qualify Better Leads
- Gartner — How to Use the BANT Framework to Qualify SaaS Leads
- SaaS Hero — Data-Driven Enterprise B2B Lead Scoring Guide
- Monday.com — Lead Scoring Rules: Prioritise Leads and Boost Sales
- Salesmotion — BANT Sales Framework: The Complete Guide
- Vanderbuild — What is B2B Lead Scoring: The Ultimate Guide
- Only B2B — B2B Lead Scoring Best Practices